Lesson Plan

L No	TOPICS	Course Outcome Addressed
1	Introduction to the course.	
2	Graphs, degree sequence, degree sum formula, and related problems.	CO1
3	Graphical vectors and Havel's -Hakimi algorithm.	CO1
4	The Erdos–Gallai algorithm	CO1
5	Connectedness, distance, eccentricity, and related results.	CO1
6	Graph Isomorphism, complement graphs	CO1
7	Self-complementary graphs and their construction	CO1
8	Trees	CO2
9	Fundamental results on Trees.	CO2
10	Traversability : Eulerian graphs	CO2
11	Hamiltonian graphs and related theorems	CO2
12	Connectivity: cut set, bridge, and related theorems	CO2
13	Whitney's Theorem and the construction of graphs	CO3
14	Vertex Covering and Vertex independence number of a graph	CO3
15	Edge covering and Edge independence number of a graph	CO3
16	Gallai's result on covering and independence number	CO3
17	Planar graphs, Euler's Formulae.	CO3
18	Coloring in graphs	CO3
19	The chromatic number of a graph G	CO3
20	Five coloring theorem.	CO3
21	Matrix representation of graphs	CO4
22	Adjacency matrix of a graphs and its properties	CO4
23	Bipartite graphs and its adjacency eigenvalues	CO4
23	Incidence matrix of a graph	CO4
24	Laplacian matrix of a graph	CO4
25	Problems	CO4
26	Matrix tree theorem	CO4
27	Cycle matrix	CO4
28	Path matrix	CO4
29	Shortest path algorithms	CO5
30	Dijkstra's Algorithm to find the shortest path	CO5
31	Floyd–Warshall algorithm to find the shortest path	CO5
32	Problems	CO5
33	Algorithms on the Minimal spanning tree	CO5
34	minimum spanning trees using Prim's algorithm	CO5
35	minimum spanning trees using Kruskal's algorithm	CO5
36	More Problems	CO5

References:

Text Books:

1. F. Harary, *Graph theory*, Narosa Publishers, 1988.
2. Narsingh Deo, *Graph theory with applications to Engineering and Computer Science*, Prentice Hall, 1987.
3. Robin J.Wilson, *Introduction to Graph theory*, Logman, 1985

Reference Books:

1. R. B. Bapat, Graphs and Matrices, Hindustan Book Agency, New Delhi, 2000.

2. D. B. West, Introduction to Graph Theory. Prentice Hall, 2001.

3. J. A. Bondy, U. S. R. Murty, Graph Theory, Springer, 2008.

Submitted by:
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(Signature of the faculty)

Date: 10-01-2026

Approved by:

PO	PO Statements
PO1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Engineering Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO11	**Life-long learning**: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.